RETAIL PROJECT (AI & ML)

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->**Importing retail_price.csv**

```
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving retail_price.csv to retail_price.csv
```

->**Loads retail pricing data from a CSV file into a pandas DataFrame and prints the top rows to preview the content**

```
import pandas as pd
import io

df = pd.read_csv(io.BytesIO(uploaded['retail_price.csv']))
print(df.head())  # Display the first few rows
```

```
  product_id product_category_name  month_year  qty  total_price  \
0       bed1        bed_bath_table  01-05-2017    1        45.95
1       bed1        bed_bath_table  01-06-2017    3       137.85
2       bed1        bed_bath_table  01-07-2017    6       275.70
3       bed1        bed_bath_table  01-08-2017    4       183.80
4       bed1        bed_bath_table  01-09-2017    2        91.90

   freight_price  unit_price  product_name_lenght  product_description_lenght  \
0      15.100000       45.95                   39                         161
1      12.933333       45.95                   39                         161
2      14.840000       45.95                   39                         161
3      14.287500       45.95                   39                         161
4      15.100000       45.95                   39                         161

   product_photos_qty  ...  comp_1  ps1        fp1      comp_2  ps2  \
0                   2  ...    89.9  3.9  15.011897  215.000000  4.4
1                   2  ...    89.9  3.9  14.769216  209.000000  4.4
2                   2  ...    89.9  3.9  13.993833  205.000000  4.4
3                   2  ...    89.9  3.9  14.656757  199.509804  4.4
4                   2  ...    89.9  3.9  18.776522  163.398710  4.4

          fp2  comp_3  ps3        fp3  lag_price
0   8.760000   45.95  4.0  15.100000      45.90
1  21.322000   45.95  4.0  12.933333      45.95
2  22.195932   45.95  4.0  14.840000      45.95
3  19.412885   45.95  4.0  14.287500      45.95
4  24.324687   45.95  4.0  15.100000      45.95

[5 rows x 30 columns]
```

->**To clean the data and analyze profitability by comparing the product's price with competitors' prices.**

```
# Check for missing values
print(df.isnull().sum())

# Drop rows with missing critical fields
df.dropna(subset=['qty', 'unit_price', 'total_price', 'freight_price', 'comp_1', 'comp_2', 'comp_3'], inplace=True)

# Estimate Cost of Goods Sold (COGS) as a proxy: unit_price - avg competitor price
df['avg_comp_price'] = df[['comp_1', 'comp_2', 'comp_3']].mean(axis=1)
df['estimated_margin'] = df['unit_price'] - df['avg_comp_price']
df['profit'] = df['qty'] * df['estimated_margin']
```

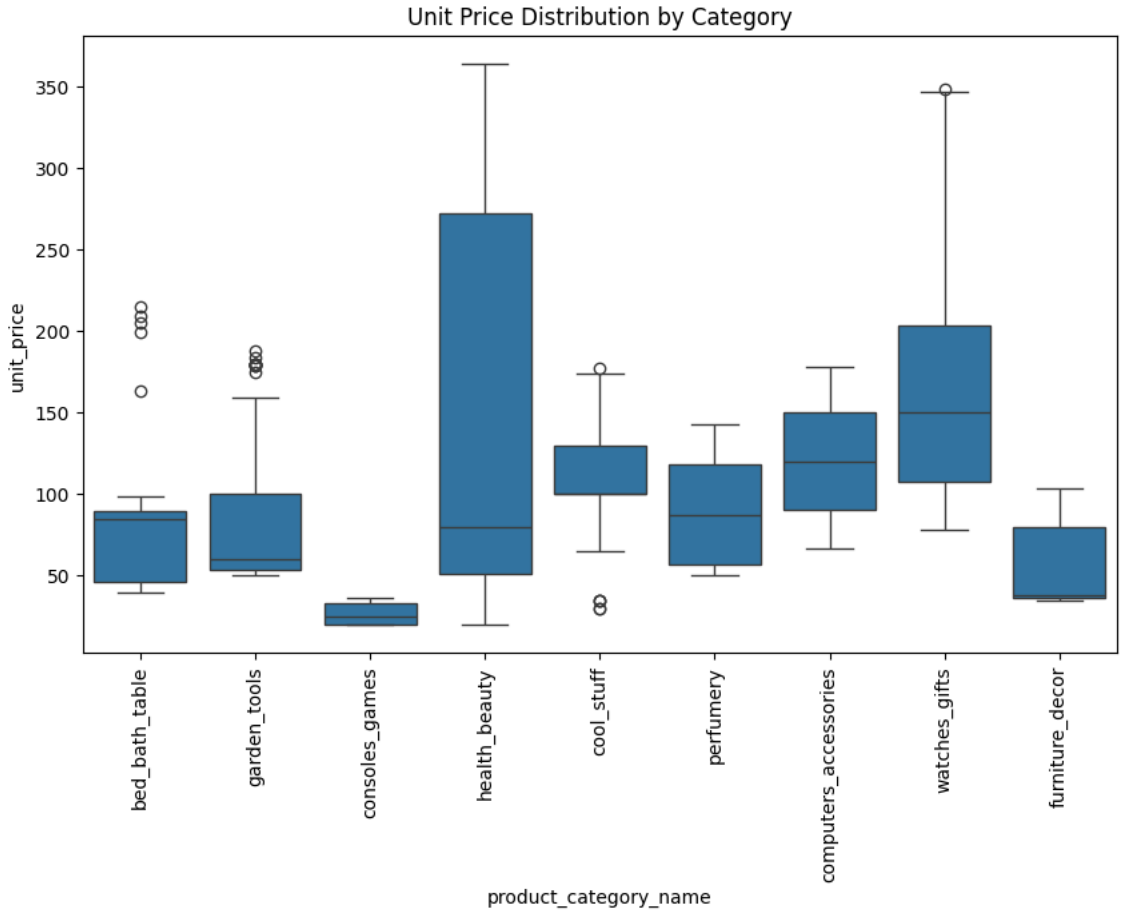
```
product_id                    0
product_category_name         0
month_year                    0
```

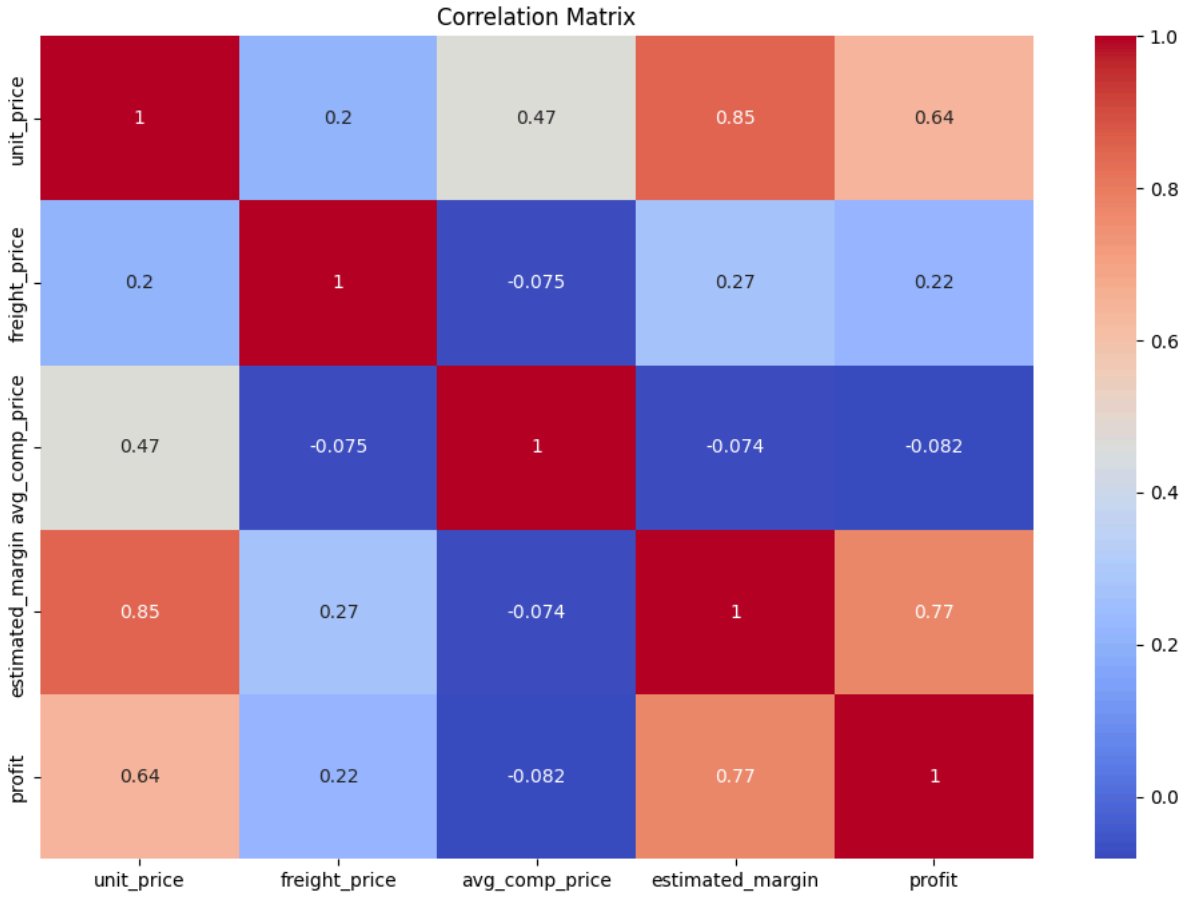
```
qty                           0
total_price                   0
freight_price                 0
unit_price                    0
product_name_lenght           0
product_description_lenght    0
product_photos_qty            0
product_weight_g              0
product_score                 0
customers                     0
weekday                       0
weekend                       0
holiday                       0
month                         0
year                          0
s                             0
volume                        0
comp_1                        0
ps1                           0
fp1                           0
comp_2                        0
ps2                           0
fp2                           0
comp_3                        0
ps3                           0
fp3                           0
lag_price                     0
dtype: int64
```

->**To visualize pricing trends across categories and explore relationships between key business metrics like cost, margin, and profit.**

```
# Plot the distribution of unit prices by product category
plt.figure(figsize=(10, 6))
sns.boxplot(x='product_category_name', y='unit_price', data=df)
plt.xticks(rotation=90)
plt.title('Unit Price Distribution by Category')
plt.show()

# Correlation heatmap
plt.figure(figsize=(12, 8))
corr = df[['unit_price', 'freight_price', 'avg_comp_price', 'estimated_margin', 'profit']].corr()
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```





->To analyze the pricing impact on sales and profit for a specific product, helping identify the optimal price point for maximizing revenue or profit.

```
# Group by product and price to see aggregated performance
agg = df.groupby(['product_id', 'unit_price']).agg({
    'qty': 'sum',
    'profit': 'sum'
```